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EDA:

- . plotted the histogram of the sellinprice column to understand the overall price distribution of vehicles in the dataset.
- .correlation matrix plotted to identify which numerical feature is important for predicting vehicle selling price
- .plotting boxplot between year and selling price to observe yearwise affect of the resale value
- .plotting scatter plot of car usages intensity with selling price to see how car_usages affect the selling price
- .barplot of grouping transmission and body type is plotted to examine which body type have high selling price ,which transmission(automatic,manual) have high price

DATA CLEANING :

- .categorical columns were converted into lowercase and extra space is removed
- .vechicle_age feature is created because it represents depreciatiton more effectively than manufacturing year ,because older vechiles have lower selling price
- .creating low and high mileage feature because these features help in analyzing vechicle condititon and depriciation pattern
- .missing numerical values were handled using median imputation and duplicate records are removed
- .outlier detectiton is performed using IQR method ,extreme selling price values were removed to reduce the noise

.condition_category feature is created by grouping vehicle condition values into low, medium, and high categories because vehicle having better condition has high selling price

.condition_odometer_ratio feature is created to capture the relationship between usage and vehicle and usage condition

.odometer_bucket feature is created to divide mileage values into low, medium, high because vehicles in higher mileage show lower selling price

MODEL TRAINING & HYPERPARAMETER TUNING:

.categorical and numerical features were selected and preprocessing applied on it

.target variable is defined and data split into training and testing part

.LightGBM Regressor was selected for model training because our dataset is large and contains too many features so it is better to choose LightGBM which captures nonlinear pattern

.Hyperparameter tuning was performed using RandomizedsearchCV on LightGBM model by testing multiple combinations of parameters. A 3-fold cross-validation strategy with MAPE scoring was used to identify the best-parameter

.The tuned LightGBM model performs better than randomforest. The MAPE improved from 20% to 13.74% after hyperparameter tuning